Question 1 (1 mark)

Which of the following should take place during the requirements definition step when developing a software solution?

- Identifying algorithms to be used
- Documenting security vulnerabilities
- Identifying potential security threats in the software
- Documenting the needs and constraints of the software

Answer:

Documenting the needs and constraints of the software

Question 2 (1 mark)

Which of the following best describes SMTP?

- A protocol for email transmission
- A set of rules to securely transfer files across a network
- A set of rules for securing communication between computers
- A protocol used for synchronising communication across networks

Answer:

A protocol for email transmission

Question 3 (1 mark)

A company conducts all of its consultations with its clients via its website.

Which of the following is the best way for the company to maintain the security of its clients' data?

- Implement data encryption
- Ensure that the software is user-friendly
- Focus on software that incorporates metadata
- Use software that allows for easy data integration with other platforms

Answer:
Implement data encryption

Question 4 (1 mark)

Which of the following are benefits of collaborating with others when developing software solutions?
(More than one may be selected.)

- ☐ Privacy is improved.
- ☐ Security of data is increased.
- ☐ Intellectual property is protected.
- ☐ Different points of view can be considered.
- ☐ Tasks can be delegated to those with expertise.

Answer:

Different points of view can be considered.

Tasks can be delegated to those with expertise.

Question 5 (1 mark)

New systems can be implemented in a number of different ways.
Match the implementation methods to their characteristics by dragging them to their corresponding boxes.

:: Direct | :: Parallel | :: Phased | :: Pilot

	The new system runs immediately with no delay.
	The new system gradually replaces the old system.
	The new system and the old system run at the same time.
	The new system is trialled with a small group of users and then implemented entirely when ready.

Answer:

:: Direct	The new system runs immediately with no delay.
:: Phased	The new system gradually replaces the old system.
:: Parallel	The new system and the old system run at the same time.
:: Pilot	The new system is trialled with a small group of users and then implemented entirely when ready.

Question 6 (1 mark)

A company is creating a new online application.

Which of the following is the least important consideration when choosing an open-source front-end web framework for the new application?

- Popularity of the framework on social media
- Quality of existing framework documentation
- Knowledge of the framework within the company's development team
- Compatibility of the framework with the company's existing technology

Answer:

Popularity of the framework on social media

Question 7 (1 mark)

Consider the following Python code.

```
for index in range(n):
    print(index)
```

Select the correct option to complete the equivalent pseudocode.

```
FOR index = [ ]
    Display index
NEXT index
```

0 TO n
1 TO n
0 TO n-1
1 TO n-1

Answer:

```
FOR index = [0 TO n-1]
    Display index
NEXT index
```

Question 8 (2 marks)

The strategies listed below can be used to test the security of a web application that contains a database.

Classify each strategy as EITHER static application security testing (SAST) OR dynamic application security testing (DAST).

	Static Application Security Testing (SAST)	Dynamic Application Security Testing (DAST)
Simulate attacks on the web application	○	○
Check the way SQL queries are constructed	○	○
Test the database's web interface for vulnerabilities	○	○
Test the web application used to interact with the database when it is running	○	○
Analyse the code for connecting to the database to identify security vulnerabilities	○	○

Answer:

	Static Application Security Testing (SAST)	Dynamic Application Security Testing (DAST)
Simulate attacks on the web application	○	●
Check the way SQL queries are constructed	●	○
Test the database's web interface for vulnerabilities	○	●
Test the web application used to interact with the database when it is running	○	●
Analyse the code for connecting to the database to identify security vulnerabilities	●	○

Question 9 (2 marks)

Place the following steps of a number guessing game algorithm in the correct order by dragging the steps into position.

:: Display "Guess a number between 0 and 100:"

:: Display "Well Done"

:: Set number = random integer between 0 and 100

:: ENDWHILE

:: WHILE guess <> number

:: Get guess

BEGIN GuessingGame

Set guess = -1

END GuessingGame

Answer:

BEGIN GuessingGame

Set guess = −1

:: Set number = random integer between 0 and 100

:: WHILE guess <> number

:: Display "Guess a number between 0 and 100:"

:: Get guess

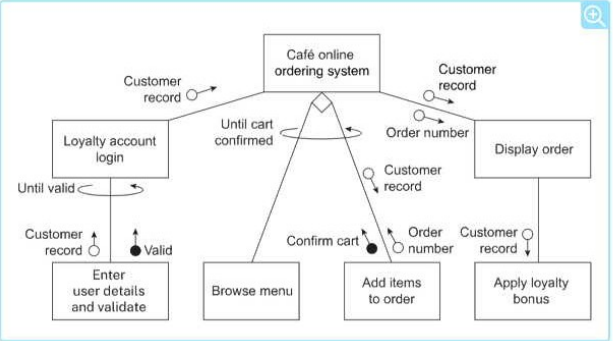
:: ENDWHILE

:: Display "Well Done"

END GuessingGame

Question 10 (2 marks)

This structure chart describes an online ordering system for a café.



Which of the following statements are TRUE about this structure chart?
(More than one may be selected.)

- **Confirm cart** is Boolean.
- Order number is a control variable.
- A valid loyalty account is not required for online ordering.
- The order cannot be displayed until the cart is confirmed.
- A choice is repeatedly offered between **Browse menu** and **Add items to order**.

Answer:

- [x] **Confirm cart** is Boolean.
- [ ] Order number is a control variable.
- [ ] A valid loyalty account is not required for online ordering.
- [x] The order cannot be displayed until the cart is confirmed.
- [x] A choice is repeatedly offered between **Browse menu** and **Add items to order**.

Question 11 (3 marks)

Consider the following algorithm.

```
BEGIN responsiveScreen
     Get ScreenSize
     IF ScreenSize >= 1024 THEN
          Viewport = 'Desktop'
     ELSEIF ScreenSize >= 768 THEN
          Viewport = 'Tablet'
     ELSE
          Viewport = 'Phone'
     ENDIF
     Display "Display set to ", Viewport
END responsiveScreen
```

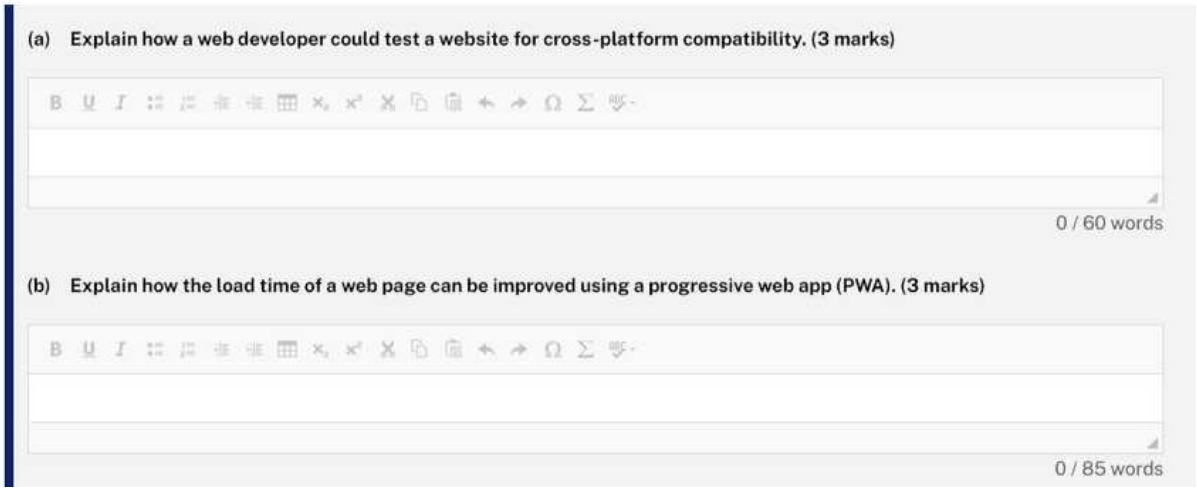
Provide a set of test data that will thoroughly test the algorithm. Include the expected outputs and reasons for inclusion. You may assume that all input data are valid.

Test data (ScreenSize)	Expected output (Viewport)	Reason for inclusion

Sample answer:

Test data (ScreenSize)	Expected output (Viewport)	Reason for inclusion
1024	Desktop	Boundary check
768	Tablet	Boundary check
100	Phone	Below lowest screen size check
850	Tablet	Between Tablet and Desktop
2000	Desktop	Above highest screen size check

Question 12 (6 marks)



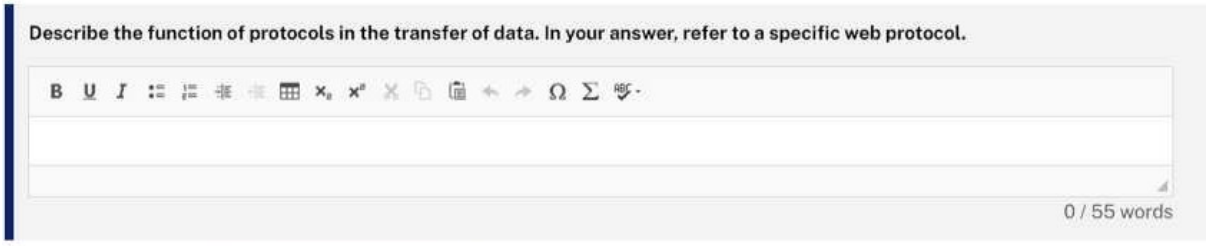
(a) Sample answer:

A web developer can use browser developer tools. These tools, built into most modern browsers, allow developers to simulate how their website looks and behaves on different devices and screen sizes. By using these tools, developers can identify and fix any issues related to cross-platform compatibility, ensuring that the website works well across a variety of devices and browsers.

(b) Sample answer:

A progressive web app (PWA) can optimise assets by compressing files and pre-caching important resources, allowing files such as image-gallery.js and video-gallery.js to be loaded into memory or cache while the rest of the page loads. PWAs also use service workers to cache resources, ensuring faster load times on repeat visits and offline functionality. They also use lazy loading, where only the necessary content is loaded initially, improving load speed, while other non-essential resources are loaded as needed.

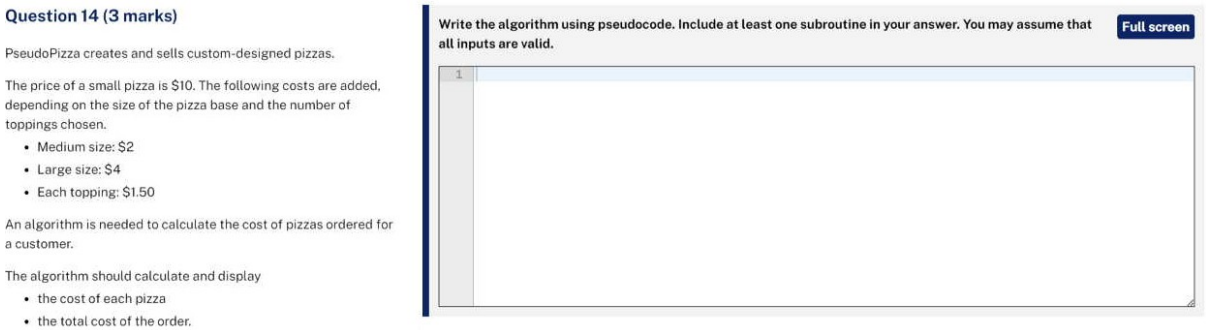
Question 13 (3 marks)



Sample answer:

Protocols are important in data transfer as they establish rules for communication between devices connected to a network, such as the internet. HTTP (Hypertext Transfer Protocol) enables browsers to request web content from servers, which respond with the requested data. The protocol defines the structure of requests and the response so that data transfer can occur.

Answers could also include TCP/IP, DNS, FTP, SFTP, SSL, TLS, SMTP, POP3, IMAP and Data packets.



Question 14 (3 marks)

PseudoPizza creates and sells custom-designed pizzas.

The price of a small pizza is $10. The following costs are added, depending on the size of the pizza base and the number of toppings chosen.

- Medium size: $2
- Large size: $4
- Each topping: $1.50

An algorithm is needed to calculate the cost of pizzas ordered for a customer.

The algorithm should calculate and display

- the cost of each pizza
- the total cost of the order.

Write the algorithm using pseudocode. Include at least one subroutine in your answer. You may assume that all inputs are valid.

Full screen

1

Sample answer:

```
BEGIN main
      Get num_pizzas
      total_cost = 0
      FOR i = 1 TO num_pizzas
            Get pizza_size

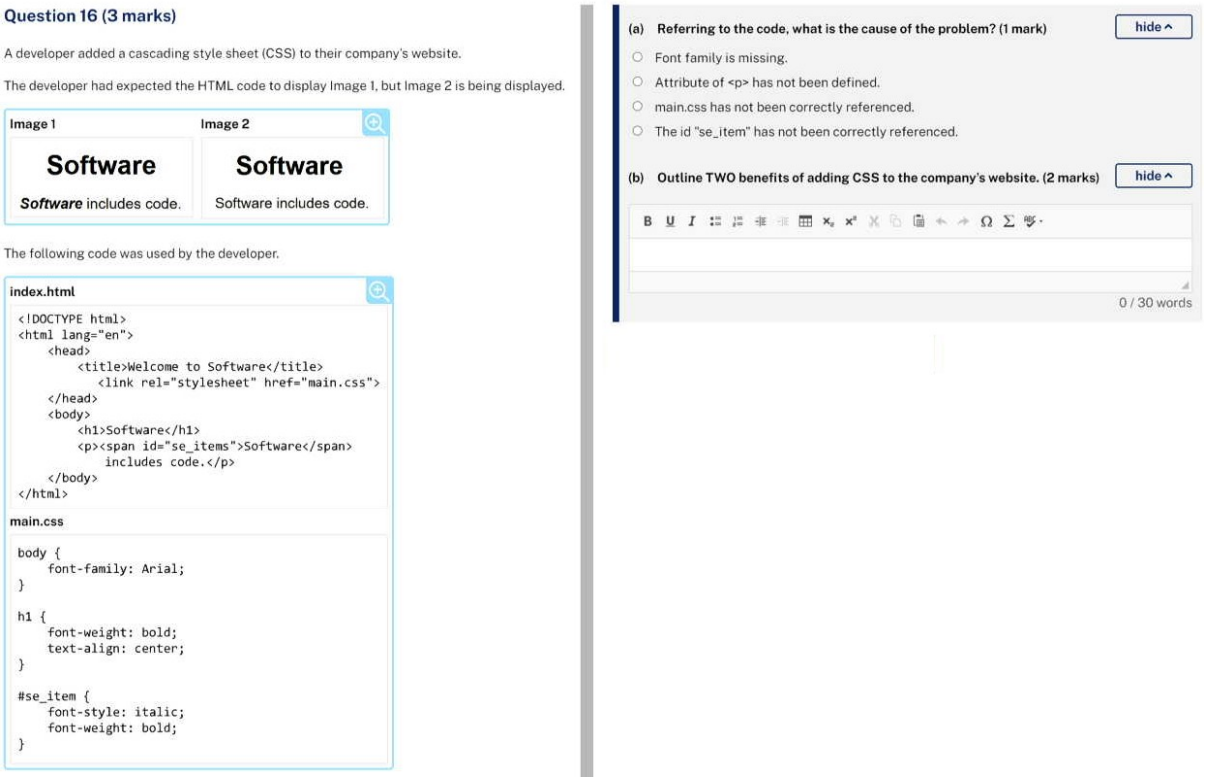
            Get num_toppings
            cost_per_pizza = calculate_pizza_cost(pizza_size,
      num_toppings)
            DISPLAY cost_per_pizza
            total_cost = total_cost + cost_per_pizza
      NEXT i
      DISPLAY total_cost
END main

BEGIN calculate_pizza_cost(size, num_toppings)
      base_price = 10
      topping_price = 1.5
      IF size = "Medium" THEN
            base_price = base_price + 2
      ELSEIF size = "Large" THEN
            base_price = base_price + 4
      ENDIF
      pizza_cost = base_price + num_toppings * topping_price
      RETURN pizza_cost
END calculate_pizza_cost
```



Sample answer:

Machine learning algorithms can be trained with a subset of the data to automatically identify patterns and trends within large datasets. The algorithms can handle complex relationships and nonlinearities present in the data allowing for efficient and comprehensive analysis. Machine learning can also be used for predictions and forecasting. These models can learn from new data over time, continuously improving their analysis and predictive accuracy and performance.



(a) Answer:

The id "se_item" has not been correctly referenced.

(b) Sample answer:

It can provide the same look and feel across the company's websites for consistency and ease of use. Separate html and CSS code can allow developers to work independently on files.

Question 17 (3 marks)

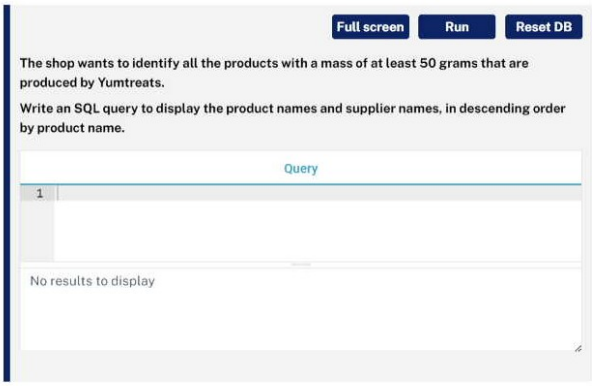
A shop uses an online database for recording its stock. The database contains a Products table and a Suppliers table. The prices are in dollars and the masses are in grams.

Products

ProductID	Name	Price	Mass	Producer	Quantity	SupplierID
PID001	TruffleSway	1.00	30	Yumtreats	100	SID001
PID002	PeanutPop	1.00	35	Yumtreats	500	SID002
PID003	Carameluxe	2.20	50	Yumtreats	100	SID003
PID004	BerryBlend	1.25	50	Yumtreats	8	SID001
PID005	Chocolate Cascade	1.25	52	Yumtreats	50	SID002
PID006	Hazelnut Symphony	2.50	50	Sugaryum	0	SID003
PID007	Caramel Delight	2.50	40	Treatopia	20	SID002
PID008	ChocoGlow	2.50	55	Treatopia	95	SID002

Suppliers

SupplierID	Name
SID001	The Lolly Pop
SID002	Sugar Rush Hour
SID003	Sweet Retreat



Sample answer:

```
SELECT Products.Name AS 'Product', Suppliers.Name AS 'Supplier'
FROM Products, Suppliers
WHERE Products.SupplierID = Suppliers.SupplierID
    AND Products.Producer = 'Yumtreats'
    AND Products.Mass >= 50
ORDER BY Products.Name DESC
```

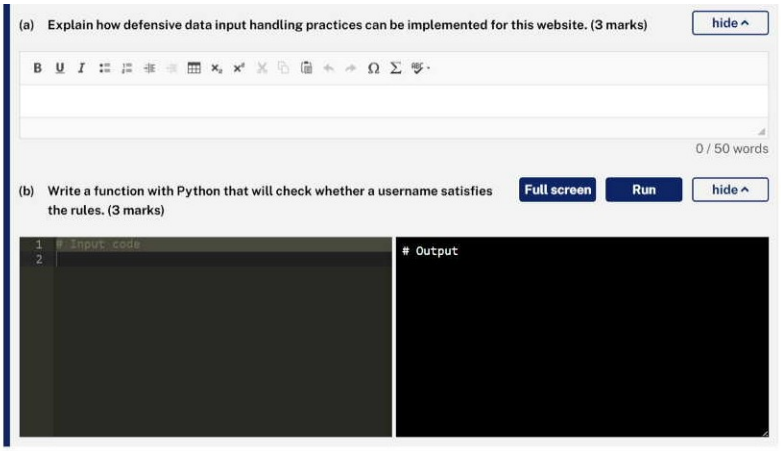
Question 18 (6 marks)

An online site requires its users to create accounts. These accounts are created using a 'Create Profile' web form. This web form includes fields such as username, password, first name, last name, mobile number and date of birth.

Each username must satisfy the following rules:

1. It must be at least 8 characters long.
2. It must start with an uppercase letter.
3. The second last character must be a lowercase letter.
4. The last character must be numeric.

An example of a valid username is NESA24s3.



(a) Sample answer:

Data sanitisation using client-side scripting can clean the data and remove dangerous input. Input validation on the client-side can check that entered data meets all of the requirements before getting processed by the server. Errors generated by data not meeting rules can be shown to the user.

(b) Sample answer:

```
def validate_username(username):
      valid = True

      # length check
      if len(username) < 8:
            valid = False
      elif (not username[0].isupper()):
            valid = False
      elif (not username[-2].islower()):
            valid = False
      elif (not username[-1].isdigit()):
            valid = False

      return valid
```

Question 19 (4 marks)

An app is being developed for use in a school. It will allow teachers to upload and assign work, and track student homework. Students can download work and upload their completed responses for marking.

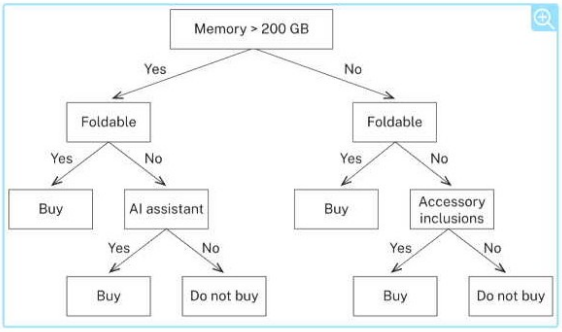
Explain how authentication and authorisation could be applied to this app.

Sample answer:

Authentication ensures that only staff and students with valid usernames and passwords, which should be encrypted in storage (eg using hashing), can access the app. Authorisation controls access to specific features of the app based on user roles or groups. Teachers may have permissions to upload, assign and track homework, while students can only download tasks and upload completed work. By organising users into specific roles or groups, the app restricts access to appropriate functionality, ensuring that each user can only perform actions relevant to their role or group.

Question 20 (4 marks)

Consider the following decision tree of a trained machine learning model that determines whether to purchase a mobile phone.



(a) Using the decision tree, determine the outcome of each of the following situations. (1 mark) hide

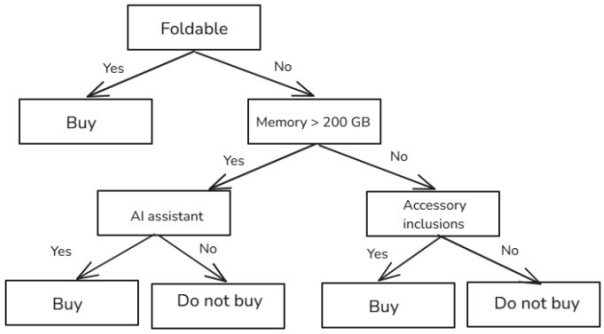
	Buy	Do not buy
Memory = 256 GB, AI Assistant = Yes, Accessory inclusions = No	○	○
Memory = 128 GB, AI Assistant = No, Accessory inclusions = Yes	○	○
Foldable = Yes, AI Assistant = No, Accessory inclusions = No	○	○

(b) The decision tree can be simplified without compromising its logic. Redraw the decision tree to reduce the number of branches. (3 marks) Full screen Reset hide

(a) Answer:

	Buy	Do not buy
Memory = 256 GB, AI Assistant = Yes, Accessory inclusions = No	◉	○
Memory = 128 GB, AI Assistant = No, Accessory inclusions = Yes	◉	○
Foldable = Yes, AI Assistant = No, Accessory inclusions = No	◉	○

(b) Sample answer:



Question 21 (6 marks)

At a school, each student studies five subjects. The school timetable has five periods each day, and each period is allocated to a different subject. The subjects shift to a different period each day so that each subject will appear in all five periods during the week.

An example of a student timetable is shown.

Period	Monday	Tuesday	Wednesday	Thursday	Friday
1	English	Sport	Science	Drama	Software
2	Software	English	Sport	Science	Drama
3	Drama	Software	English	Sport	Science
4	Science	Drama	Software	English	Sport
5	Sport	Science	Drama	Software	English

In this example, the student studies English, Software, Drama, Science and Sport. From Monday to Friday, English is held in Period 1, Period 2, Period 3, Period 4 and Period 5. Software is held in Period 2, Period 3, Period 4, Period 5 and then Period 1. A similar pattern can be observed for the other subjects.

You are required to write a Python program to print out the timetable in the following format:

```
Monday
    Period 1: English
    Period 2: Software
    Period 3: Drama
    Period 4: Science
    Period 5: Sport
```

and so on ...

Start your program with: `subjects = ["English", "Software", "Drama", "Science", "Sport"]`

Different students may study different subjects.

The program will be used by different students to print out their timetable by changing the subjects listed in the first line of code.

(a) If the timetable is stored as a 2D array, what is the correct Python snippet for accessing the subject that is on Monday Period 3? (1 mark) hide

- `Timetable[2]`
- `Timetable[1,4]`
- `Timetable[0][2]`
- `Timetable[1].Subject`

(b) Write the Python program using subroutines. (5 marks) Full screen Run hide

```
1 # Input code
2
```

```
# Output
```

(a) Answer:

Timetable[0][2]

(b) Sample answer:

```
subjects = ["English", "Software", "Drama", "Science", "Art"]
NUM_DAYS = 5
NUM_PERIODS = 5
DAYS_OF_WEEK = ["Monday", "Tuesday", "Wednesday", "Thursday",
"Friday"]

def create_timetable(subjects):
    timetable = []
    for day in range(NUM_DAYS):
        periods = []
        for period in range(NUM_PERIODS):
            periods.append(subjects[(period - day) % NUM_PERIODS])
        timetable.append(periods)
    return timetable

def print_timetable(timetable):
    print()
    for day in range(NUM_DAYS):
        print(f"{DAYS_OF_WEEK[day]}")
        for period in range(NUM_PERIODS):
            print(f"\tPeriod {period+1}: {timetable[day][period]}")
        print()


timetable = create_timetable(subjects)
print_timetable(timetable)
```

Question 22 (5 marks)

An online relational database is used to keep track of students at a coding club. The contents of the database are shown.

Students

StudentID	FirstName	Surname	Attended	Level
Student1	John	Doe	5	Beginner
Student2	Jane	Smith	4	Intermediate
Student3	David	Kim	3	Beginner
Student4	Sarah	Lee	2	Expert
Student5	Emma	Wilson	1	Beginner
Student6	Michael	Johnson	0	Intermediate
Student7	Olivia	Davis	2	Beginner
Student8	Ethan	Martinez	3	Intermediate
Student9	Sophia	Choi	4	Expert
Student10	Noah	Moore	5	Beginner
Student11	Isabella	Garcia	1	Intermediate
Student12	Liam	Vo	2	Expert

(a) Complete the SQL query to generate the number of students at each level. (2 marks) hide

SELECT Level, [ ]

FROM Students

[ ] BY [ ]

(b) Race conditions have been identified as a potential issue for this database. hide

Provide an example of when a race condition may occur in this scenario and outline how secure code could be implemented to prevent it. (3 marks)

0 / 50 words

(a) Answer:

SELECT Level, COUNT(Level)

FROM Students

GROUP BY Level

(b) Sample answer:

With multiple teachers potentially accessing the database at the same time, race conditions can allow malicious users to gain unauthorised access. Authentication could be used to restrict access to the database, and code can be written to use atomic transactions and database locks to time the reading and writing of data.

Question 23 (6 marks)



(a) Sample answer:

Training data is initially selected by humans, who may unintentionally introduce their own biases. To minimise this, ensure that training data is representative of the problem the ML model will address. Reviewing data by individuals from diverse backgrounds could identify and reduce potential biases, ensuring fairer and accurate model performance.

(b) Sample answer:

Linear regression and K-nearest neighbours (KNN) are both supervised learning algorithms that make predictions from labelled data. The key difference is how they handle data: linear regression assumes a linear relationship between input features and the target variable, while KNN makes predictions based on the closest ‘K’ training examples, allowing it to handle more complex, non-linear patterns. However, KNN can be more computationally expensive than linear regression and is usually slower.

Question 24 (4 marks)

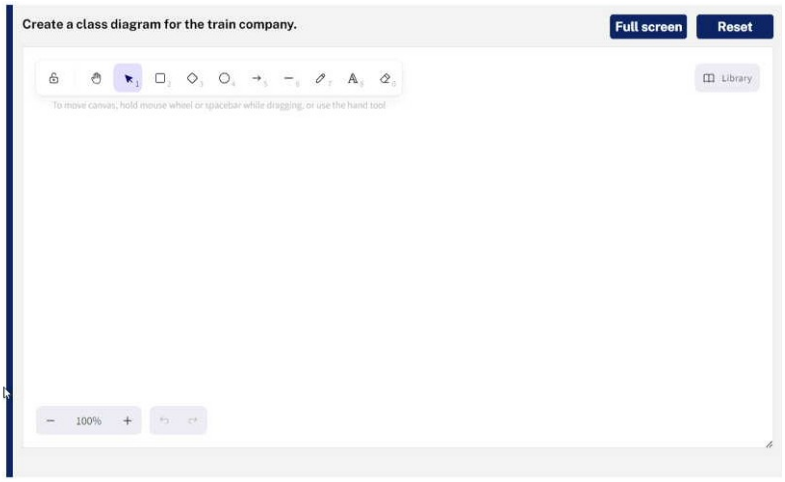
A train company wants a class diagram that models the relationship between long-distance train trips and passengers.

The long-distance train has:

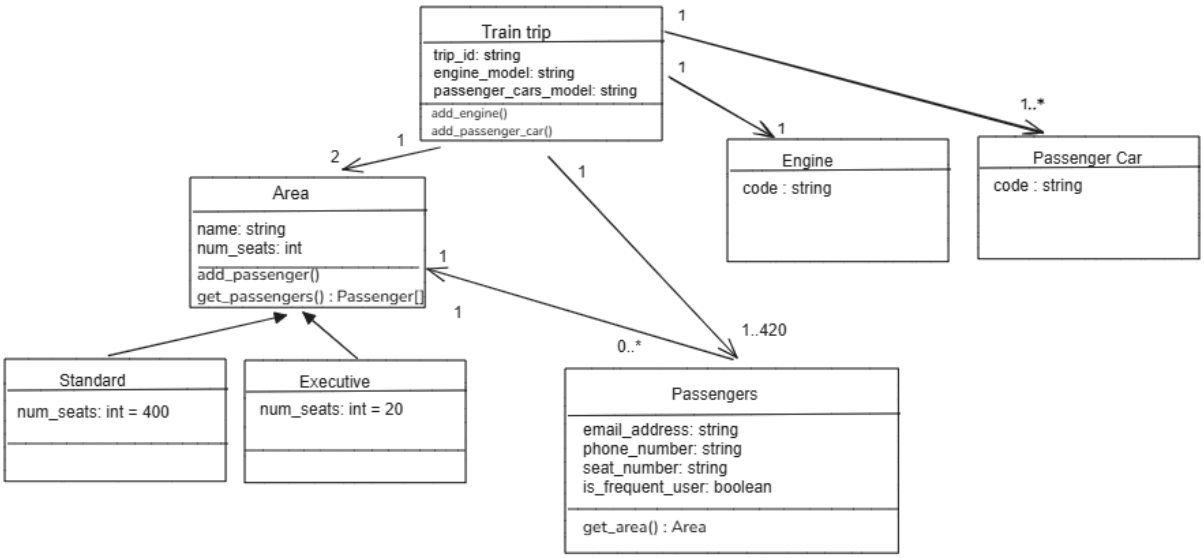
- a unique trip identifier (eg AL456)
- a model code for the train engine (eg E67) and a model code (eg P9754) for each of the passenger cars
- two different areas ('Standard' and 'Executive'). The Executive area has 20 seats and there are 400 seats in the Standard area.

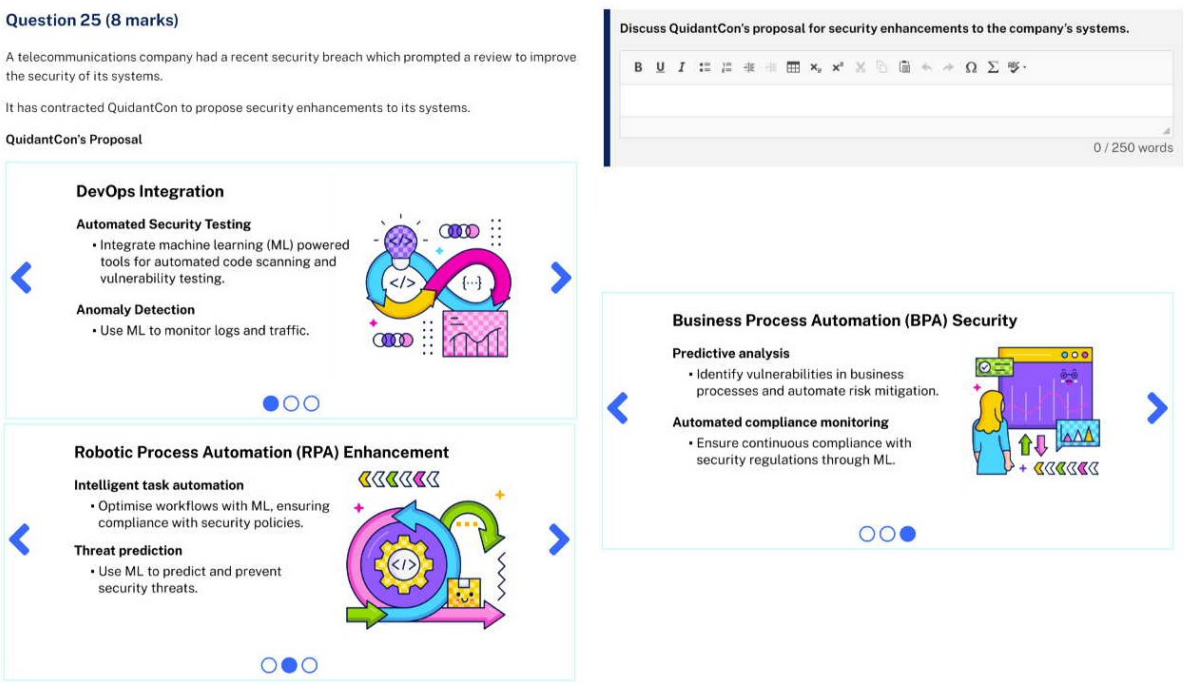
Each passenger:

- has contact details including an email address and phone number
- purchases a 'Standard' or 'Executive' ticket and is given a seat number (eg 25D)
- may be a frequent traveller.



Sample answer:





Sample answer:

QuidantCon's proposal for the company suggests leveraging machine learning (ML) and automation to bolster security measures. One key benefit is the automation of security testing and vulnerability scanning, facilitated by integrating ML-powered tools into the DevOps pipeline. This approach ensures continuous security testing and provides rapid feedback to developers, allowing for early detection of security issues and reducing the risk of vulnerabilities being exploited. The frequent and automated testing improves the overall security position and enhances compliance with industry standards. Additionally, ML algorithms can analyse vast amounts of data to detect patterns and anomalies, which is crucial for proactive threat identification. This capability enables the company to identify security risks proactively, learn from past incidents, and predict future attacks, thereby enhancing their overall security measures.

QuidantCon's proposal for the company also raises some concerns. One key issue is the need for high-quality data for ML algorithms to work effectively. If the data is inconsistent or incomplete, the ML models may not perform well, leading to false positives or missed threats. Another concern is the complexity of implementing ML solutions, which can be challenging and may require specialised skills and training. This complexity could result in integration challenges, higher costs, and delays. Additionally, ML systems can be biased, potentially affecting security decisions. Security and privacy are also critical, as handling sensitive data requires protection against unauthorised access and breaches. These concerns emphasise the need for a careful and balanced approach when considering ML and automation in security practices.